

Dawson SPARKS: Secondary Particle Angular Radiation from Kosmic-Rays using Spark-Chambers

Team Dawson SPARKS

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INTRODUCTION

Galactic Cosmic Rays (GCRs), solar particle events and solar wind are the main sources of radiation in space [1]. As space missions grow in scale and range, characterizing radiation and shielding materials for astronauts and electronic components is becoming critical [1], as seen during the Artemis I mission when the Orion spacecraft faced significant electronic disturbances [2]. GCRs consist of isotropic hydrogen nuclei (87%) [3] and are the main source of ionizing radiation. High hydrogen materials (e.g. water) are common shielding options, reducing the risk of neutron radiation and secondary showers while attenuating energy of the nuclei [5]. We want to study the angular spread of secondary radiation in these materials (mostly electrons) induced by GCRs, with the PS beam as a proxy thereof. Characterizing the angular distribution of secondary radiation is essential to determining the volumetric radiation dose behind shields for spacecraft and habitat geometries [6, 7]. Experimental validation is critical to calibrating Monte Carlo simulations (e.g. GEANT4) used to generate radiation heatmaps.

EXPERIMENT PROPOSAL

The experimental setup (see figure 1) would first consist of a 10x10x10cm water target, or other GCR shielding material like a regolith replicate. The spark chamber will be placed at different positions behind the target to characterize the angular scattering distribution of the emitted secondary radiation. To simulate GCRs (typically 10^{-3} GeV to 10^3 GeV [3] with maximum flux around 1 GeV), we would use the PS beam from 3.5-15 GeV [4]. We expect these parameters to generate an average of 13 primaries in the 500 ns recombination window of our spark chamber assuming spills of 10^4 particles.

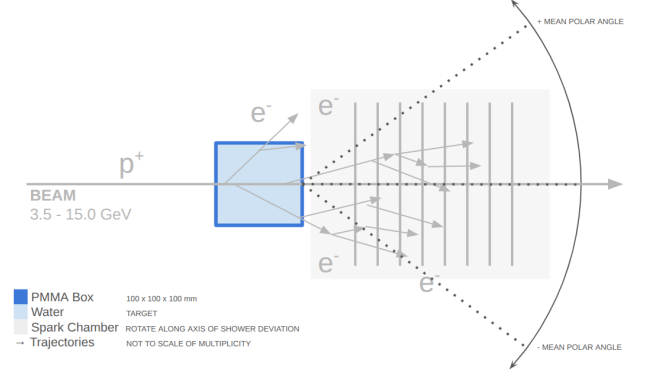


FIG. 1. Representation of our proposed experimental setup. The proton beam passes through the water target in front of the spark chamber. (Showers not to scale with multiplicity of p^+ from 3.5-15 GeV). The spark chamber can be rotated and deviated from the beam to capture different angular distributions, with the simulated mean polar angles represented.

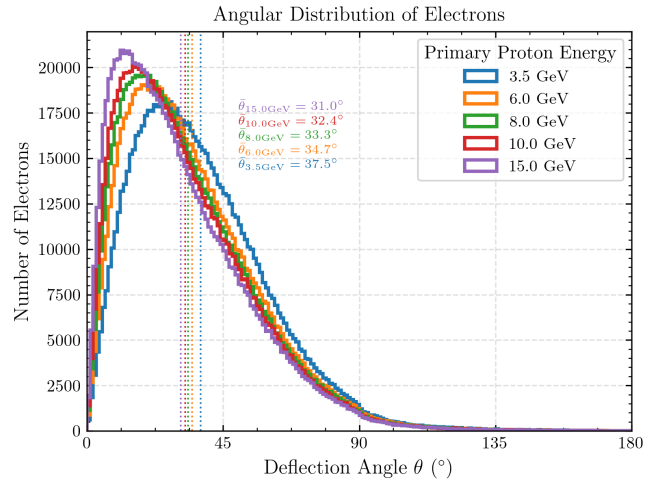


FIG. 2. Plot is derived from 10 million primary proton events per energy level using GEANT4 (FTFP_BERT)[8, 9]. 10 million protons were fired at the center of a one-liter cube of water with 2mm thick plexiglass (PMMA) wall at varying energy levels. The number of electrons ($n = 917277$) produced at 3.5 GeV were used at each energy level to make a linearized version of the graph. Details in Table I.

TABLE I. Secondary Electron Angular Distributions from GEANT4, Confirming Highland Distribution from Multiple Coulomb Scattering

Energy (GeV)	Mean Angle °	RMS Angle °	FWHM	Multiplicity
3.5	37.523	43.432	48.6	0.09173
6.0	34.736	40.934	45.0	0.13009
8.0	33.35	39.725	43.92	0.15281
10.07	32.428	38.937	42.48	0.17066
15.07	31.027	37.789	39.6	0.20206

Our time at the beamline would be divided into

1. Calibrating the spark chamber with an angular sweep through the beam at a constant energy. During this step, we would also find the ideal collimator settings for our experiment, ideally the smallest one available for positional calibration.
2. Testing the position and angular ranges for a water target for different beam energies.
3. Testing the position and angular ranges for a regolith replicate target (time permitting).

SPARK CHAMBER DESIGN

Hardware

The spark chamber that we conceptualized and built is triggered by polyvinyl toluene (PVT) plates through silicon-photomultiplier (SiPM) photon detection. Student-designed SiPM biasing and amplifier boards [10] are discriminated for coincidence by our FPGA analog data acquisition system [11], given a selected threshold.

High voltage is generated by the Source Trigger Unit (STU): a student-designed, galvanically isolated, resonant flyback adjustable high-voltage supply with a user-selected trigger mechanism via argon-filled triggered spark gap (TSG), or discrete insulated gate bipolar transistors (IGBT) [12].

A flyback transformer is driven at resonance with an adjustable DC bias, which results in a linearly-related generation of a 0-10 kV output voltage (see figure 10). Buffer capacitors (1-2 nF) are charged for each pair of spark chamber plates through a 2-10 M Ω output impedance.

Upon the digital coincidence given by the FPGA, an exciter coil resonates the trigger pin of the TSG with a 3 kV pulse, or isolated gate drivers pulse two series IGBTs—in order to achieve conduction through the selected mechanism. The previously-positive pole of buffer capacitors are now grounded; the high-voltage is applied across the plates. The electronic propagation time is <1 μ s for the TSG and <400 ns for the IGBT trigger mechanisms from SiPM impulse to voltage application (see figure 6)—acceptable considering recombination time in literature [13].

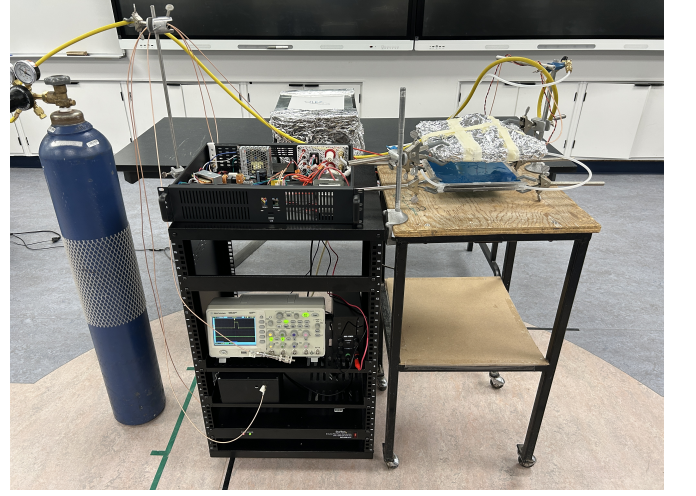


FIG. 3. Spark chamber experimental setup. From left to right. Argon cylinder (CAS 7740-37-1, >99.9%) with high pressure regulator feeding into gas lines (yellow). Trigger cart: source trigger unit (STU) receiving trigger from data acquisition system (aluminium box) with oscilloscope on coincidence and spark timing. Testing cart: trigger PVT plate with SiPM wrapped in aluminium. Three aluminium plates under (two effective sparks) to confirm working principle.

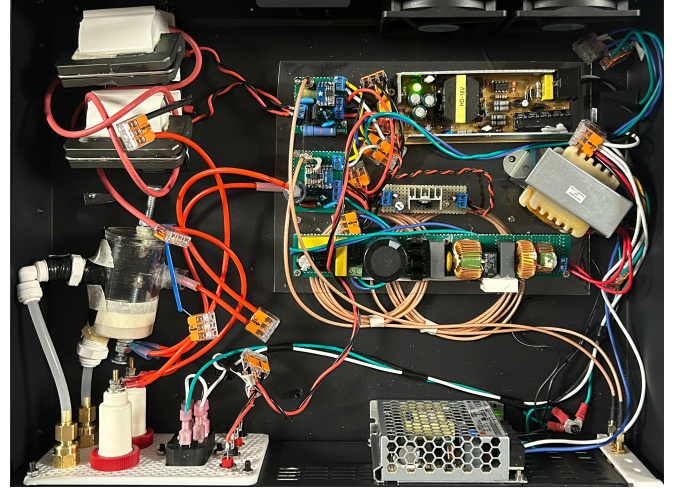


FIG. 4. Source trigger unit (STU): adjustable high voltage power supply and trigger generator. Top left: dual-core, series high-voltage flyback transformer. Bottom left: power input/output and triggered spark gap with gas lines. Top right: trigger and power electronics [12]. Bottom right: resonant clock input for high-voltage generation and trigger line.

Argon serves as the dielectric for its ease of obtention; 4-6 kV is necessary to spark plates separated by a gap of 8 mm, mechanically supported by laser-cut acrylic panels for airtightness, electrical insulation, and visual clarity.

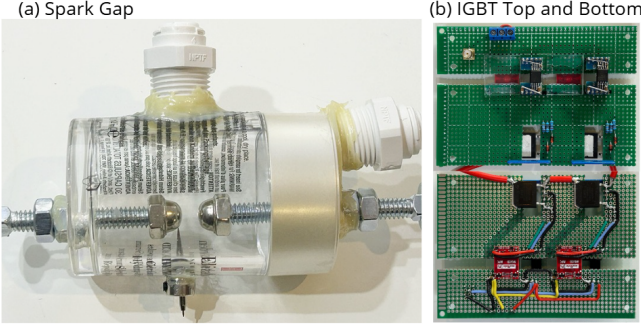


FIG. 5. (a) Current prototype triggered spark gap with 6.35 mm dome nuts as principal adjustable contacts, and sharpened needle as trigger electrode. 6.35 mm NPT tube press fittings are used for argon gas intake and electrode flushing (top), and exit (right). (b) Spark gap replacement in implementation: series insulated gate bipolar transistor (IGBT) trigger. Isolation channels manually carved in PCB. Top view: 8 kV isolated gate drivers (1ED3123MC12H), tuned gate drive resistors and static balancing resistors. Bottom view: isolated gate drive power supplies (MGJ2D121505MPC) with 4.5 kV IGBTs in series (IXYT30N450HV).

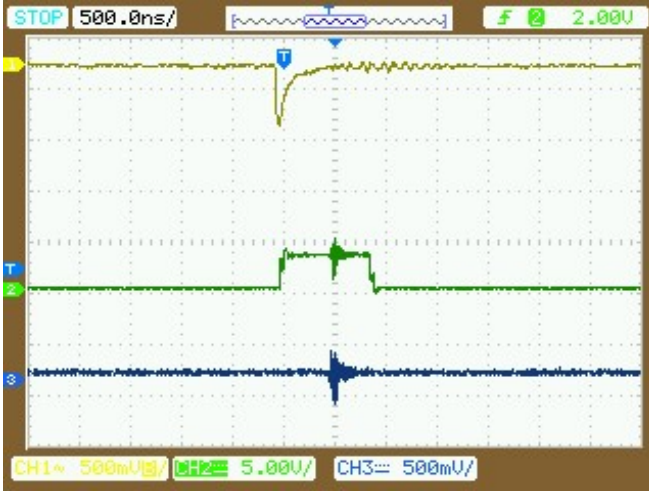


FIG. 6. Triggered spark gap oscilloscope trace. Yellow: PVT (Eljen EJ-200) scintillation on SiPM (OnSemi 60035) after transimpedance amplification (OPA858). Green: 3V3 CMOS, 1µs duration, active-high coincidence output from the FPGA. Blue: near E-field interference from spark gap breakdown in argon. Propagation delay $<1\mu\text{s}$.

Acquisition

High-framerate fisheye cameras and electret microphones detect spark positions and are corroborated through software. Three fisheye cameras onlook the active region of the spark chamber, solving for perspective-n-point (PnP) via predetermined reference ArUco markers [14], and detecting sparks optically. Additionally, at least four microphones are positioned on the top and bottom of the spark chamber, allowing for layered-

multilateration, and single 3D track reconstruction, assuming linearity of particle trajectory. [15]

Data Analysis

We localize the lines of intersection from Time Differences of Arrival (TDoA) composites from the microphone array, where the reconstructed spark locations form a compact cluster near its theoretical geometric center. By choosing one reference microphone, we select a subset of $(n - 1)$ TDoA's, each providing us with a hyperbolic restriction. Using the least squares sum method, we reconstruct the 3D path.

Machine Learning Model

We addressed the possibility of spark discharges happening at very short intervals with a change of neural-network model architecture; instead of using a simple multilayer perceptron, we treat the problem as a set assignment problem. Our model [16] is based on the DETector TRansformer (DETR) [17]. We flattened the 2D convolutional architecture for images developed by META to work with 1D time-domain signals from our microphones (see figure 7).

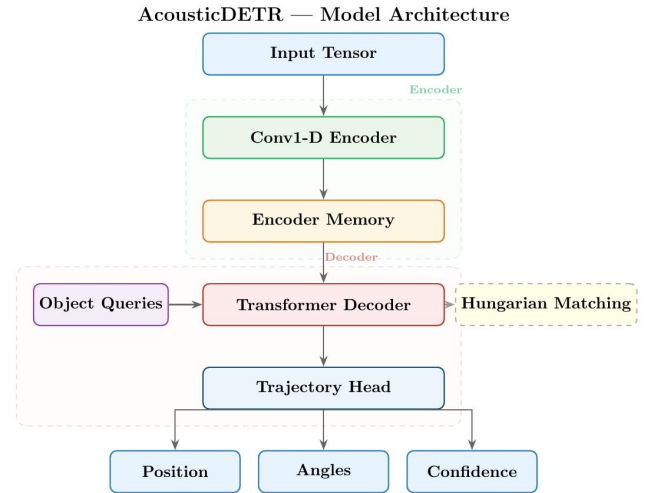


FIG. 7. A n plates by m microphones by 256 longitudinal amplitude samples tensor is inputted into the model, which outputs up to K track predictions with a position, and a sigmoidal confidence score optimized on binary cross-entropy loss. A convolutional pass kernels signal timing and amplitude into a fixed-sized tensor, attended to by a transformer encoder-decoder. Sound signals from separate sparks summed into the processed audio signal may therefore be decomposed into individual position predictions.

AcousticDETR — 3-track pileup event

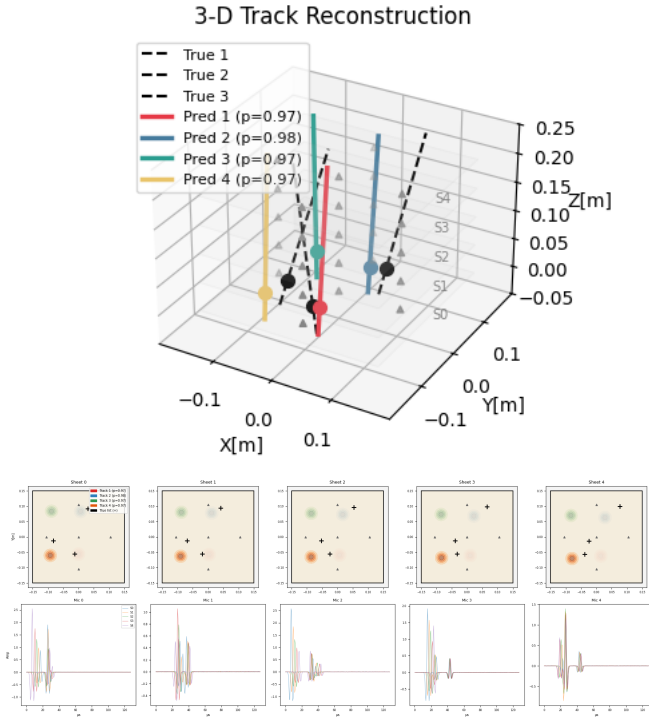


FIG. 8. Trained on 8000 simulated events sectioned into 100 epochs, the figure shows an example of the 3-pileup track. It includes a 3-D path reconstruction, our four queries ($K = 4$) on each sheet (spark chamber layer) and the amplitude of a microphone per sheet. For higher resolutions, refer to AcousticDETR [16].

LIMITATIONS

Due to the low rate-tolerance of the spark chamber as of writing, a high rate of protons might be problematic. We suggest three potential solutions. First, we might reduce the momentum of the beam, decreasing electron multiplicity (see Table I). Second, we suggest the use of collimators to restrict the beam, thus reducing the particle flux. Third, considering the radiation length of protons and electrons in water, a decrease in the length of the target can reduce showers and thus electron flux.

Furthermore, at specific GeV ranges, significant quantities of pions are created, partially decaying into muons. It will be impossible to completely filter them without also blocking the desired protons. As such, we propose recording background runs, which will allow us to determine the expected noise levels.

CONCLUSION

From this experience, we hope to learn about the atmosphere of a real research environment and to give back to our community. As a group, we have collaborated with a variety of people, from fellow students and faculty to external academic institutions, and we hope to collect data on GCRs to give back to the scientific community. As we worked towards our proposal, we bridged the gaps of our varying skill levels and united different fields of science such as physics, computer science, and math. For us, the prospect of visiting a world-renowned research center like CERN is a driving source of inspiration.

OUTREACH

We hope to use our skills and knowledge to give back to the community. We hope to make science available to people outside the scientific community by presenting our project at the annual Dawson Science Fest, a public event where science students share their research. We also hope to be able to use part of our funding to design and donate small, cost-effective detectors to local high schools, to help with science education and increase public interest in physics, especially among younger students. We also hope to be able to contribute and maybe even participate in upcoming conferences such as the RADiation and its Effects on Components and Systems Conference (RADECS) at the end of September 2026 which closely align with the goals outlined in this research proposal.

ACKNOWLEDGMENTS

This project would not have been possible without our professor and mentor Dr. Manuel Toharia, who provided us with background knowledge in particle physics and valuable feedback on our project. We would like to thank both him and SPACE coordinator Joel Trudeau for bringing us together in the High-Energy Physics club. Furthermore, we would like to thank the Student Academic Growth and Enrichment (SAGE) Fund for providing the financial resources required to develop our Spark Chamber. We are very grateful towards the Geant4 collaboration for providing the open-source simulation toolkit used in this study. We also acknowledge the extensive documentation and community support which facilitated our characterization of secondary radiation for space shielding. We acknowledge the use of Artificial Intelligence (AI) Tool: Claude 4.6; Software: used to assist with drafting and debugging code, which was reviewed and tested by the authors.

Finally, we would like to thank the members of the Beamline for Schools competition for giving us this opportunity and organizing the competition.

APPENDIX

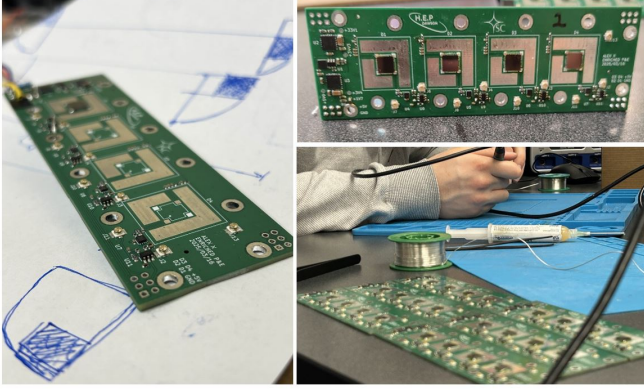


FIG. 9. Student-designed SiPM printed-circuit boards. Each PCB has four channels; a total of two PCBs are used in our detector. Each board is optically coupled to a large PVT plate. One such plate is placed on either side of the spark chamber to form a coincidence telescope. SiPMs and other components are manually populated onto the PCB.

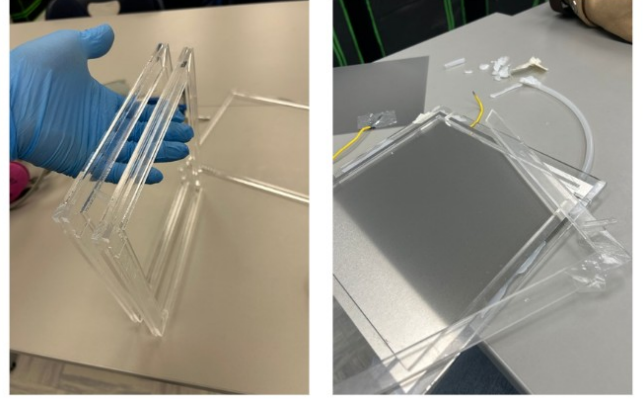


FIG. 11. Left: Acrylic separators, glued with methyl methacrylate solvent, for voltage isolation between aluminium plates to mitigate edge sparking. Right: 6061 aluminium alloy of size $200 \times 200 \times 0.8$ mm with acrylic isolators, creating 8 mm of effective gap distance between plates.

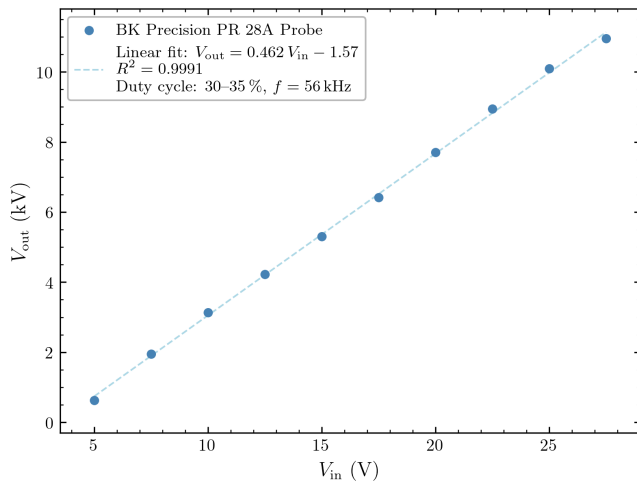


FIG. 10. STU adjustable high voltage generation. Primary-side flyback converter topology output linearity with adjustable input bias voltage.

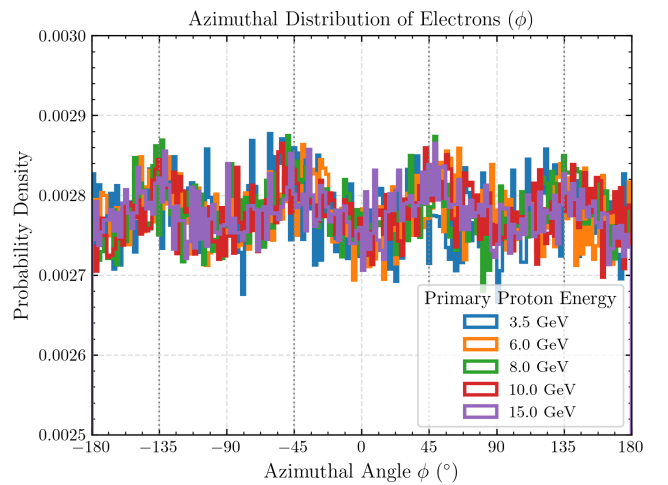


FIG. 12. The azimuthal angular distribution of electrons produced in the same simulation as figure 2 demonstrates azimuthal symmetry at each energy level. Small statistical fluctuations are due to the sample size of 10 million primaries per energy level. Nevertheless, peaks of probability density at 45° intervals align with corners of the square-based prism target, where shower rate is increased due to geometry.

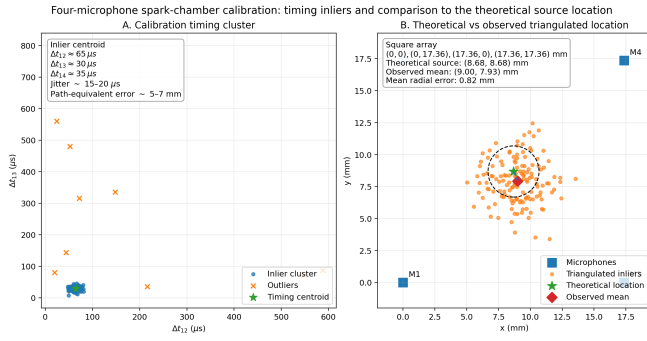


FIG. 13. Demonstration of the acoustic trilateration system localizing spark events within millimeter-scale precision under controlled calibration conditions. We observe clustering of data points near the theoretically calculated trilateration point.

* Teacher

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